

Modelado de rendimientos financieros: volatilidad condicional, colas pesadas y dependencia multivariada

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Agenda

- 1 Objective and Data
- 2 Empirical Characteristics of Returns
- 3 Volatility Modeling
- 4 Shock Modeling
- 5 Forecasting and Evaluation
- 6 Copula Multivariate Model
- 7 Conclusion

- ▶ Provide a coherent, step-by-step **recipe** for financial return modeling at the undergraduate level.
- ▶ Follow one consistent path: empirical characterization, univariate modeling, forecast evaluation, and multivariate extension.
- ▶ Keep a dual lens throughout:
 - **Statistical validity** of model assumptions.
 - **Risk-management usefulness** (VaR/ES forecasts).

Data (Training set)

- ▶ Daily returns: S&P 500 and 10-year U.S. Treasury note index (SPBDU1BT).
- ▶ Sample used across chapters: Jan 2014–Dec 2024.

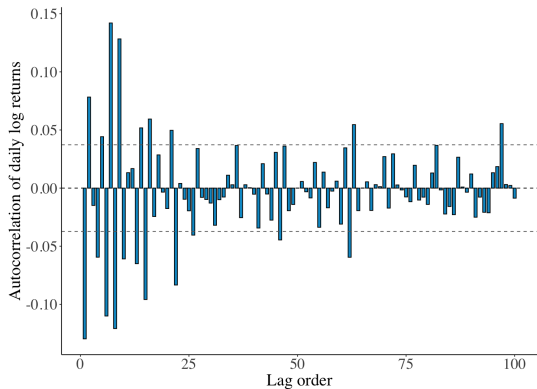
Forecast (Test set)

- ▶ Out-of-sample window: Jan 2025–Apr 2025 (81 observations).
- ▶ Forecasts: 1-day-ahead density, VaR and ES (2.5%).
- ▶ Tests: Christoffersen tests, PIT diagnostics, KS, and DM comparison.

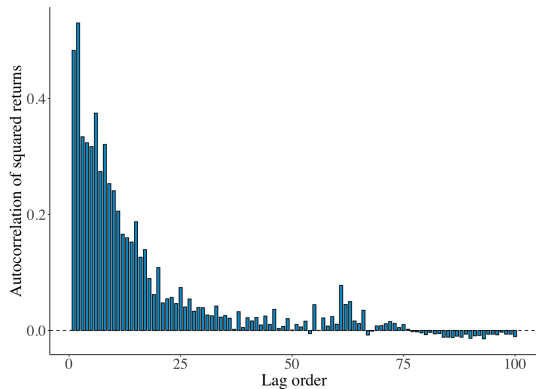
Empirical Characteristics of Returns

- ▶ Stylized facts on S&P 500 daily returns (2014–2024):
 - Near-zero autocorrelation in raw returns.
 - Volatility clustering in squared returns.
 - Leverage effect.
 - Heavy tails, asymmetry, and high kurtosis.
 - Time-varying cross-asset dependence.
- ▶ **These empirical characteristics serve as guidelines for what models should capture and replicate**

Autocorrelation Patterns



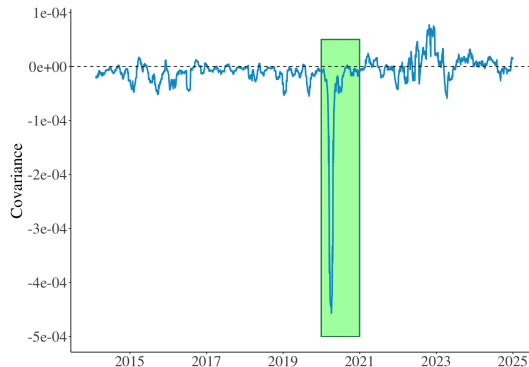
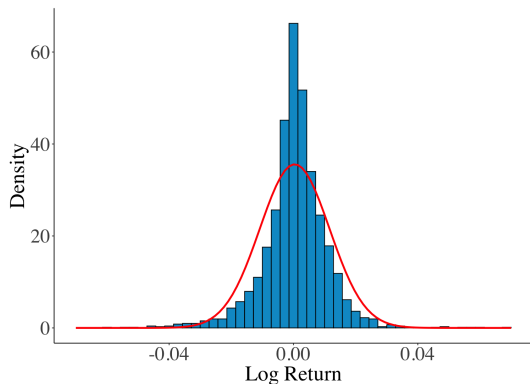
Returns ACF



Squared returns ACF

- ▶ Returns: little linear predictability.
- ▶ Squared returns: clear dependence and clustering.

Distributional and Dependence Features



- ▶ The empirical distribution exhibits heavier tails than the normal distribution and asymmetry.
- ▶ The correlation among assets appears to vary over time.

Return model

$$R_{t+1} = \sigma_{t+1} z_{t+1}, \quad z_{t+1} \sim \text{i.i.d. } D(0, 1)$$

- ▶ Goal: model conditional variance σ_{t+1}^2 .
- ▶ Ljung-Box on R_{t+1}^2 : strong evidence of conditional heteroskedasticity.
- ▶ Two models:

$$\textbf{GARCH(1,1): } \sigma_{t+1}^2 = \omega + \alpha R_t^2 + \beta \sigma_t^2$$

$$\textbf{GJR-GARCH(1,1): } \sigma_{t+1}^2 = \omega + \alpha R_t^2 + \alpha \theta I_t R_t^2 + \beta \sigma_t^2$$

- ▶ Indicator variable: $I_t = \mathbf{1}\{R_t < 0\}$.
 - If $R_t < 0$, $I_t = 1$ and shock impact is $\alpha(1 + \theta)R_t^2$.
 - If $R_t \geq 0$, $I_t = 0$ and shock impact is αR_t^2 .
- ▶ GJR term ($\theta > 0$) captures asymmetric (leverage) response.

Parameter Estimation: QMLE

- ▶ Parameters: $\vartheta = (\omega, \alpha, \beta)$ for GARCH(1,1), $\vartheta = (\omega, \alpha, \beta, \theta)$ for GJR-GARCH(1,1).
- ▶ **Quasi**-maximum likelihood: apply the normal MLE even if z_{t+1} is not normal (consistent estimates under correctly specified mean and variance).

Given a sample $\{R_t\}_{t=1}^T$, the conditional log-likelihood under normality is

$$\ell(\vartheta) = -\frac{1}{2} \sum_{t=1}^T \left[\log(2\pi) + \log(\sigma_t^2) + \frac{R_t^2}{\sigma_t^2} \right].$$

The QMLE estimator solves

$$\hat{\vartheta} = \arg \max_{\vartheta \in \Theta} \ell(\vartheta),$$

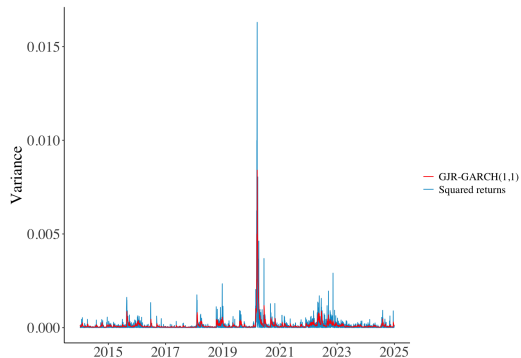
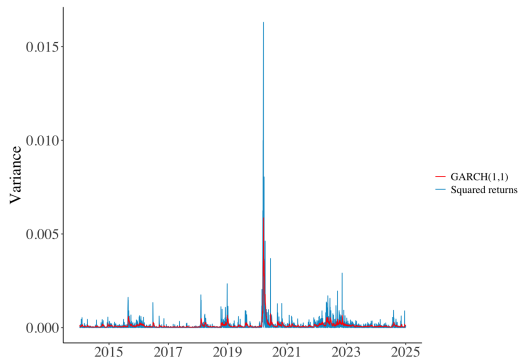
subject to positivity and finite long-run variance restrictions.

- ▶ No closed-form maximizer: solved numerically, each ϑ generating a full path $\{\sigma_t^2\}_{t=1}^T$ via the volatility recursion.

Estimated Dynamics (QMLE)

$$\text{GARCH}(1,1): \sigma_{t+1}^2 = 0.0000039 + 0.171R_t^2 + 0.794\sigma_t^2$$

$$\text{GJR-GARCH}(1,1): \sigma_{t+1}^2 = 0.0000040 + 0.054R_t^2 + 0.248I_tR_t^2 + 0.795\sigma_t^2$$

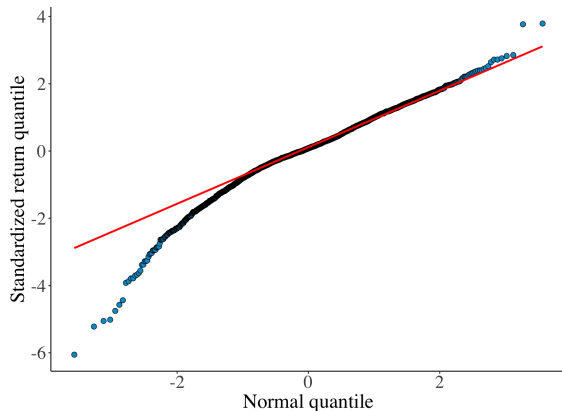
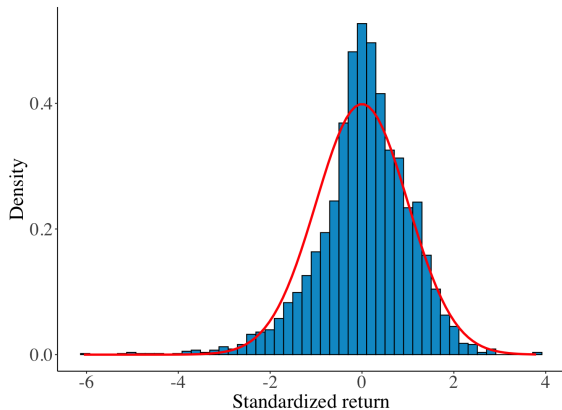


Key Takeaways on Volatility Modeling

Model	Mean QLIKE
GARCH(1,1)	1.6011
GJR-GARCH(1,1)	1.5673

- ▶ LR test for extra GJR parameter: $p < 1\%$ (Christoffersen (2012) framework).
- ▶ Both models absorb squared-return dependence.
- ▶ Persistence is close to one, consistent with financial volatility behavior.

Why Move Beyond Normal Shocks?



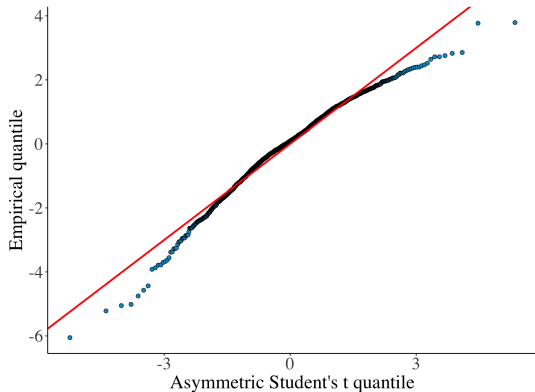
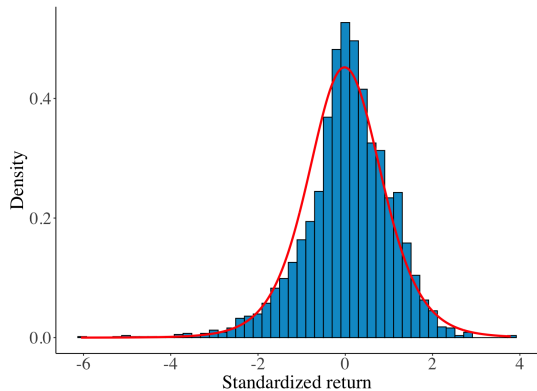
- ▶ Standardized residuals still show heavy tails and asymmetry.
- ▶ Normality is strongly rejected (very small p-values in diagnostics).
- ▶ Two alternatives considered: Asymmetric t and EVT-POT.

Approach 1: Asymmetric t Shocks

$$R_{t+1} = \sigma_{t+1} z_{t+1}, \quad z_{t+1} \sim \text{i.i.d. asyt}(d_1, d_2)$$

- ▶ Provides a skewed extension of the Student's t-distribution, enabling the modeling of both kurtosis and skewness observed in standardized returns.
- ▶ Parameters estimated after volatility QMLE step.
- ▶ Better center/tail fit than normal, but left tail remains underfit.

Asymmetric t Fit Diagnostics



- ▶ Good fit in the center and moderate tails.
- ▶ Left extreme tail remains the main weakness for risk management.

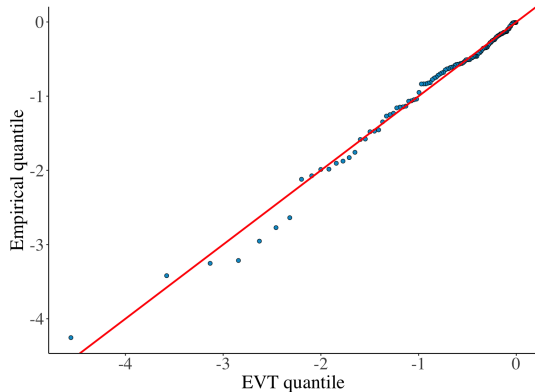
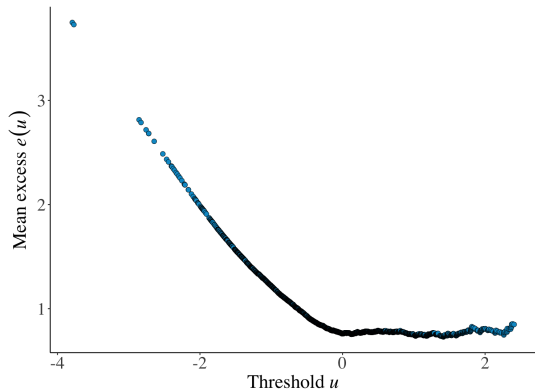
Approach 2: EVT (Peak-over-Threshold)

$$-z_{t+1} \mid (-z_{t+1} > u) \sim \text{GPD}(\xi, \beta, u), \quad u = 1.8$$

$$f(y) = \frac{1}{\beta} \left(1 + \frac{\xi(y-u)}{\beta} \right)^{-(1+\frac{1}{\xi})}, \quad y > u$$

- Focuses explicitly on left-tail risk, the critical region for VaR/ES.
- Threshold chosen through mean excess function linearity.

EVT Diagnostics



- ▶ Mean excess plot supports $u = 1.8$.
- ▶ GPD improves fit of the left tail versus full-distribution alternatives.

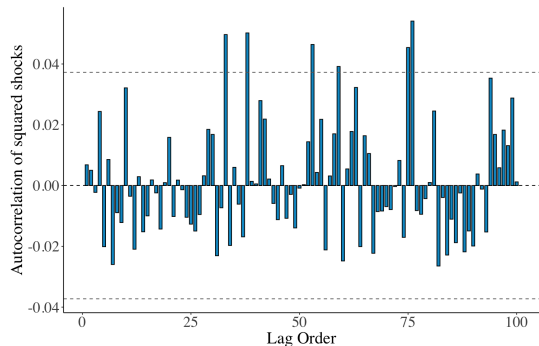
Estimated Return Models

Model	α	β	θ	Tail params
GARCH-Asyt	0.171	0.794	–	$d_1 = 7.35, d_2 = 0.0115$
GJR-Asyt	0.054	0.795	4.62	$d_1 = 7.83, d_2 = 0.0129$
GARCH-EVT	0.171	0.794	–	$\beta_{\text{GPD}} = 0.774, \xi = 0.029$
GJR-EVT	0.054	0.795	4.62	$\beta_{\text{GPD}} = 0.621, \xi = 0.206$

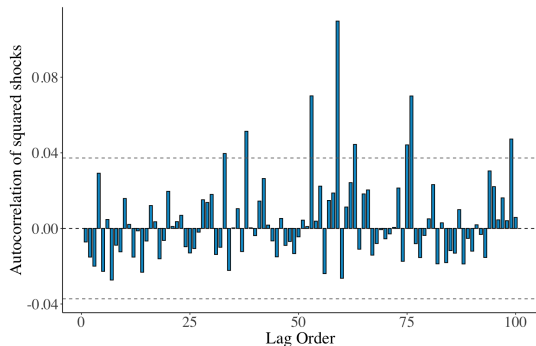
- Four complete univariate return models are ready for forecasting.

- ▶ Forecast horizon: 1 day ahead, test window Jan–Apr 2025.
- ▶ Outputs: σ_{t+1}^2 , density forecast, $\text{VaR}_{2.5\%}$, $\text{ES}_{2.5\%}$.
- ▶ Evaluation blocks:
 - Volatility diagnostics (ACF of standardized squares, QLIKE, LR).
 - Risk diagnostics (Christoffersen LR_{uc} , LR_{ind} , LR_{cc}).
 - Density diagnostics (PIT histograms, KS, Berkowitz, DM).

ACF of Standardized Squares



GARCH(1,1)



GJR-GARCH(1,1)

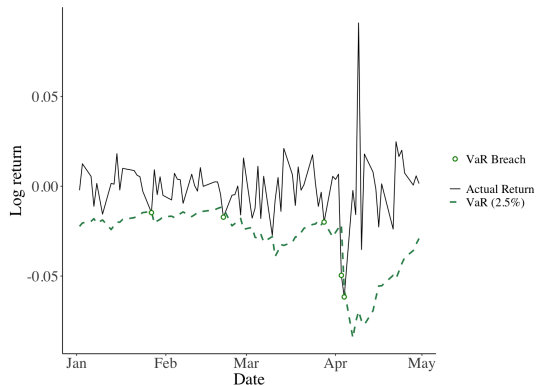
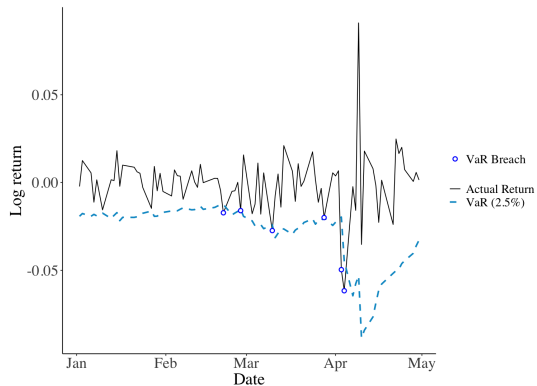
- ▶ Diagnostic goal: remove autocorrelation in standardized squared returns.
- ▶ Both models substantially absorb the dependence structure.

Asymmetric t Forecasts: VaR and ES

$$\text{VaR}_{t+1}^p = -\sigma_{t+1} F_{\text{asyt}}^{-1}(p; d_1, d_2), \quad \text{ES}_{t+1}^p = -\sigma_{t+1} \text{ES}_{\text{asyt}}(p)$$

- ▶ Full-density models provide direct VaR and ES forecasts.
- ▶ Calibration check at $p = 2.5\%$: expected violations about 2 in 81 days.

Asymmetric t VaR Breaches (Test Period)



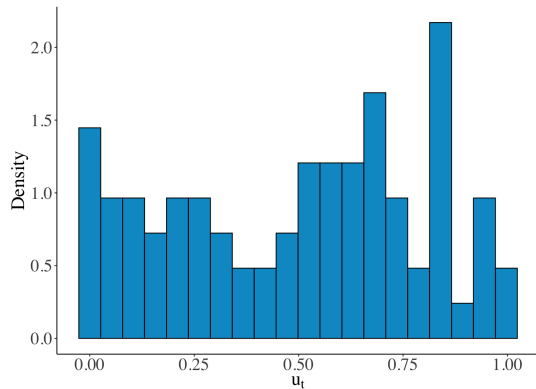
► Observed breaches exceed expectation and cluster after volatility shock.

Asymmetric t Backtesting (VaR)

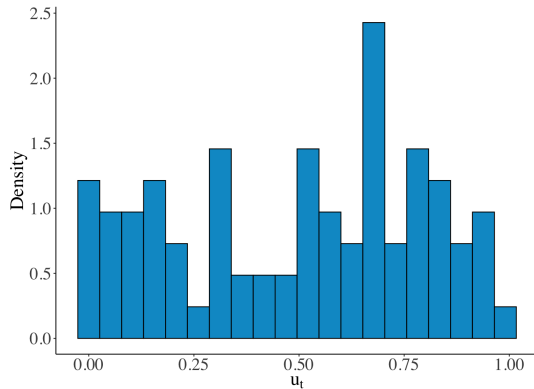
Test	p-value type	GARCH-Asyt	GJR-Asyt
LR_{uc}	Simulated	0.003	0.144
LR_{ind}	Simulated	0.095	0.047
LR_{cc}	Simulated	0.015	0.028

- ▶ VaR performance is weak: coverage and independence issues.
- ▶ Simulated LR_{uc} and LR_{cc} p-values are low for both models.
- ▶ Breaches cluster around volatility jumps, violating independence.

Asymmetric t Density Checks



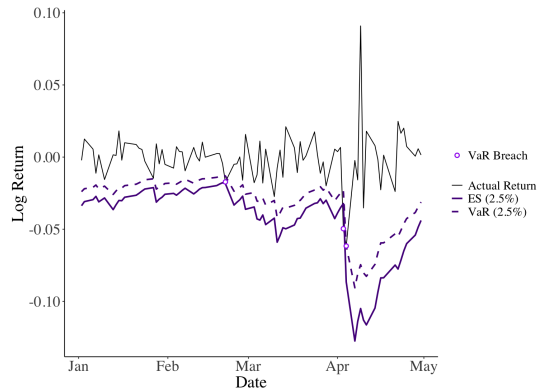
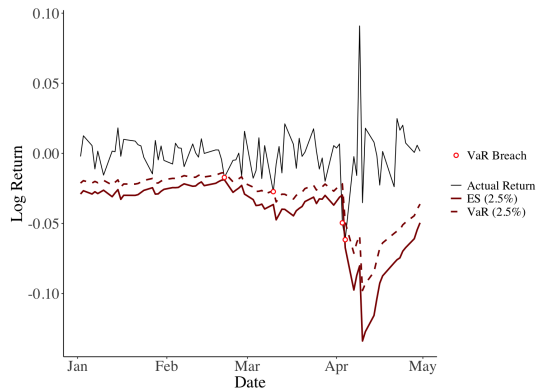
PIT histogram (GARCH-Asyt)



PIT histogram (GJR-Asyt)

► KS on $\Phi^{-1}(u_{t+1})$: 0.8512 (GARCH-Asyt), 0.7931 (GJR-Asyt).

EVT Forecasts in the Test Period



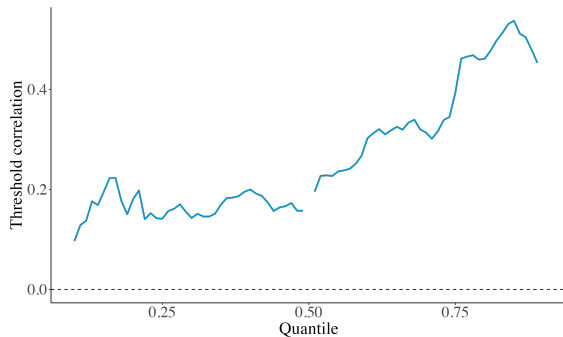
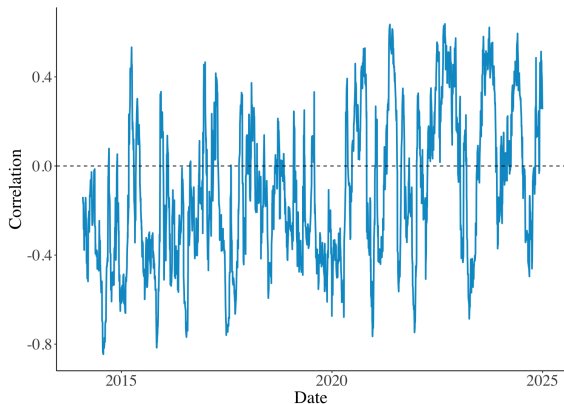
- ▶ EVT focuses directly on the left tail, giving more conservative risk forecasts.
- ▶ EVT variants show fewer breaches than Asyt counterparts.

EVT Backtesting Summary

Test	p-value type	GARCH-EVT	GJR-EVT
LR_{uc}	Simulated	0.188	0.554
LR_{ind}	Simulated	0.030	0.013
LR_{cc}	Simulated	0.217	0.200

- ▶ Coverage improves, but independence of violations remains problematic.
- ▶ Main weakness: slow adaptation to abrupt volatility regime changes.
- ▶ Stress-testing (2008 episode) supports stronger breach control for EVT variants.

Why Copulas?



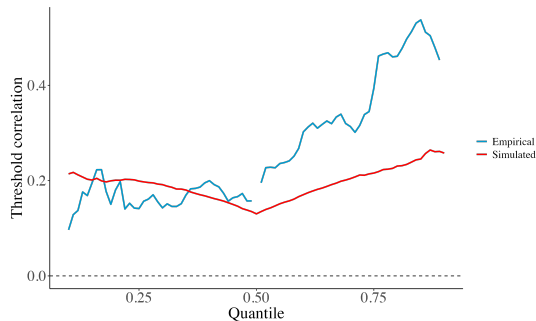
- ▶ Need dependence modeling for active risk management and diversification analysis.
- ▶ DCC captures dynamic linear dependence; copulas also target nonlinear tail dependence.
- ▶ Threshold-correlation evidence motivates copula-based modeling.

$$G(u_1, u_2; \rho^*, d) = t_{(d, \rho^*)}(t^{-1}(u_1; d), t^{-1}(u_2; d))$$

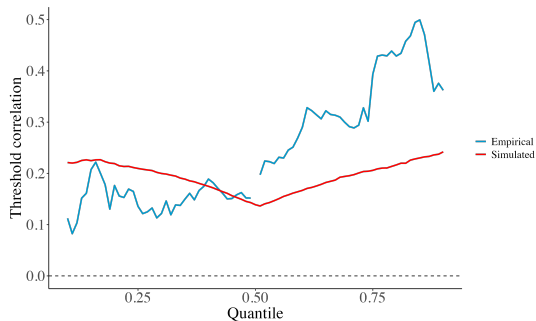
Copula model	ρ^*	d
GARCH-GARCH	-0.2121	4.9671
GJR-GJR	-0.2088	4.8304

- ▶ Static t-copula chosen for tractability and tail-thickness flexibility.
- ▶ Marginals estimated first (GARCH-Asyt / GJR-Asyt), then copula parameters.

Threshold-Correlation Fit



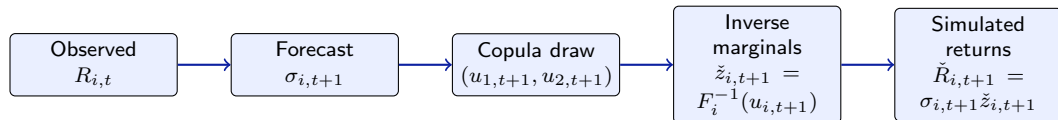
GARCH-GARCH copula



GJR-GJR copula

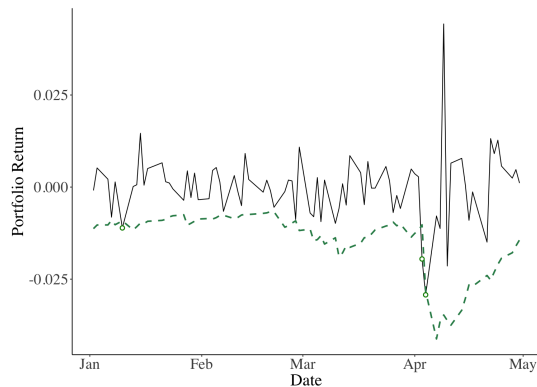
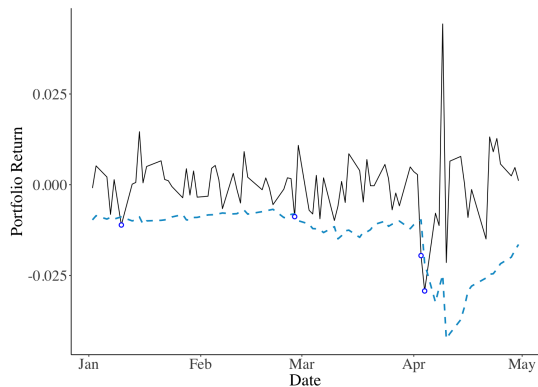
- ▶ t-copula reproduces heavier dependence in both tails reasonably well.
- ▶ Right-tail asymmetry in data is only partially captured (copula is symmetric).

Monte Carlo Forecasting Pipeline



- Repeat M times to form portfolio-return density.
- Compute VaR/ES from simulated quantiles and tail means.

Copula Portfolio Results



- Breaches are similar across the two copula variants.
- Main weakness remains clustering under abrupt volatility jumps.

Copula Backtesting and Density Validation

Test	p-value type	GARCH-GARCH	GJR-GJR
LR_{uc}	Simulated	0.188	0.554
LR_{ind}	Simulated	0.030	0.013
LR_{cc}	Simulated	0.217	0.200

- ▶ KS on normal transform: $p = 0.5271$ (GARCH-GARCH), $p = 0.4872$ (GJR-GJR).
- ▶ Density forecasts are statistically acceptable despite VaR clustering issues.

Main Conclusions

- ▶ The thesis delivers a coherent end-to-end framework from univariate to multivariate risk modeling.
- ▶ Volatility modeling: GJR improves in-sample volatility fit (QLIKE, LR evidence).
- ▶ Shock modeling:
 - Asymmetric t improves full-density fit.
 - EVT improves left-tail conservativeness for risk measures.
- ▶ Backtesting reveals a persistent challenge: **independence of violations** under sudden volatility spikes.
- ▶ Copula extension adds useful dependence structure and valid density forecasts for portfolios.

- ▶ Static dependence and symmetric copula limit adaptation and asymmetry capture.
- ▶ EVT tails improve coverage but still react slowly to abrupt regime shifts.
- ▶ Promising extensions:
 - Dynamic/asymmetric copulas (Christoffersen and Langlois (2011), Patton and Oh (2011)).
 - Shock models with time-varying skewness and kurtosis.
 - Broader stress-testing and alternative volatility proxies.

- Berkowitz, J. (2001). Testing density forecasts, with applications to risk management. *Journal of Business & Economic Statistics*, 19(4), 465–474. Retrieved from <https://doi.org/10.1198/07350010152596718> doi: 10.1198/07350010152596718
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Finance*, 31(3), 307–327. doi: [https://doi.org/10.1016/0304-4076\(86\)90063-1](https://doi.org/10.1016/0304-4076(86)90063-1)
- Bollerslev, T. (2023). The story of GARCH: A personal odyssey. *Journal of Econometrics*, 234, 96–100. doi: <https://doi.org/10.1016/j.jeconom.2023.01.015>
- Campbell, J. Y., Lo, A. W., & MacKinlay, A. C. (1997). *The econometrics of financial markets*. United States: Princeton University Press.
- Christoffersen, P. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4), 841–862. Retrieved from <https://doi.org/10.2307/2527341> doi: 10.2307/2527341
- Christoffersen, P. (2012). *Elements of financial risk management*. United States: Academic Press.

- Christoffersen, P., & Langlois, H. (2011). *The joint dynamics of equity market factors* (Working Paper No. CREATES 2011-45). CREATES, Aarhus University; University of Toronto. Retrieved from <https://doi.org/10.1017/S0022109013000598> (Later published in *JFQA* (2013), vol. 48, no. 5, pp.1371–1404) doi: 10.1017/S0022109013000598
- Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1, 223-236. Retrieved from <https://doi.org/10.1080/713665670>
- Diebold, F. X., Gunther, T. A., & Tay, A. S. (1998). Evaluating density forecasts with applications to financial risk management. *International Economic Review*, 39(4), 863–883. Retrieved from <https://doi.org/10.2307/2527342> doi: 10.2307/2527342

- Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263. Retrieved from <https://doi.org/10.1080/07350015.1995.10524501> doi: 10.1080/07350015.1995.10524501
- Fernández, C., & Steel, M. F. J. (1998). On bayesian modeling of fat tails and skewness. *Journal of the American Statistical Association*, 93(441), 359-371. Retrieved from <https://www.jstor.org/stable/2669632>
- Glosten, L. R., Jagannathan, R., & Runkle, D. E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *Journal of Econometrics*, 48(5), 1779-1801. doi: <https://doi.org/10.1111/j.1540-6261.1993.tb05128.x>

- Jondeau, E., & Rockinger, M. (2006). The copula-garch model of conditional dependencies: An international stock market application. *Journal of International Money and Finance*, 25(5), 827-853. Retrieved from <https://doi.org/10.1016/j.jimonfin.2006.04.007> doi: 10.1016/j.jimonfin.2006.04.007
- McNeil, A. J., & Frey, R. (2000). Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach. *Journal of Empirical Finance*, 7(3-4), 271-300. Retrieved from [https://doi.org/10.1016/S0927-5398\(00\)00012-8](https://doi.org/10.1016/S0927-5398(00)00012-8) doi: 10.1016/S0927-5398(00)00012-8
- Okimoto, T. (2008). New evidence of asymmetric dependence structures in international equity markets. *Journal of Financial and Quantitative Analysis*, 43(3), 787-815. Retrieved from <https://doi.org/10.1017/S0022109000004294> doi: 10.1017/S0022109000004294

- Patton, A. J. (2006). Modelling asymmetric exchange rate dependence. *International Economic Review*, 47(2), 527–556. Retrieved from <https://doi.org/10.1111/j.1468-2354.2006.00387.x> doi: 10.1111/j.1468-2354.2006.00387.x
- Patton, A. J., & Oh, D.-H. (2011). *Modelling dependence in high dimensions with factor copulas* (Working Paper). Duke University. Retrieved from https://public.econ.duke.edu/~ap172/0h_Patton_MV_factor_copula_apr12.pdf (Precursor to *JBES* 2017 paper on factor copulas)
- Tsay, R. S. (2010). *Analysis of financial time series*. United States: Wiley.

Questions

 github.com/ferpmillan/thesis

 github.com/ferpmillan/thesis-tex